2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT)
Bangalore, India. Oct 7-9, 2022

Classification of Music Genre Using Machine Learning

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***Abstract*-Deep learning approach is the first approach for classification of music genre using machine learning and for the purpose of predicting the genre label of a signal,a CNN model would be trained usually from end-to-end, spectrogram is used to carry out this process. Hand- crafted features were used in the second approach which is also the last approach. Four out of many traditional machine learning classifiers will be trained beside these features and their performance would be compared afterwards. Identification of features that would help the classification of this task needs to be performed. The databases of online music are growing day by day and nowadays it is very simple to access online music,due to this reason people are finding extremely uneasy to regulate the songs that they like to listen to. Nowadays music has become a very salient proportion of the Internet content,the most salient source of pieces of music is the net.**

Keywords: Machine Learning, Convolutional Neural Network,Music genre, Support Vector Machine

I. INTRODUCTION

In this research paper two methods will be used. The deep learning approach is the first approach of classification of music genre using machine learning and for the purpose of predicting the genre label of a signal a CNN model would be trained usually from end-to-end, spectrogram is used to carry out this process. Hand-crafted features were used in the second approach which was also the last approach. Four out of many traditional machine learning classifiers will be trained beside these features and their performance would be compared afterwards. Identification of features that would help the classification of this task needs to be performed. The databases of online music are growing day by day and nowadays it is very simple to access online music, due to this reason people are finding extremely uneasy to regulate the songs that they like to listen to. Nowadays music has become a very salient proportion of the Internet content, the most salient source of pieces of music is the net.

The various important characteristics that generally put up to make the requisite model utilized for the Music Genre Classification can be easily understood. A novel approach can be made to take out musical pattern features of the audio file with the use of CNN(Convolutional Neural Network).The various advantages of CNN in MIR (Music Information Retrieval) can be surveyed. By making use of many experiments and results, it can be concluded that CNN(Convolutional Neural Network) have the most powerful ability to catch characteristics that are informative, from the musical pattern which are varying.

Music plays a very significant role in our daily lives. Everybody loves to listen to music. Everybody is very fond of music. Music gives peace to many people. Music is usually like a remedy to the ears. We love to listen to music according to our mood that is happy,sad and angry.Music genre is an important term used in music.Music genre is basically a conventional class which is used to identify or figure out only some categories of music as belonging to a tradition that is shared tradition.There are various types of music genre as follows:pop music,UK bass,dubstep,industrial music,jungle music,electronic rock,Z-pop,latin music,disco,techno,house music,soul music,indie rock,blues,world music,country music,hip hop music,K-pop,J-pop and C-pop.There are many ways to classify music genre like using CNN(Convolutional Neural Network),deep learning techniques,python,neural networks,machine learning,image classification neural networks,KNN(k-nearest neighbour) and genetic algorithm.

II. DATASET

This table comprises of various music genre like reggae music, rock music,hip hop music, techno, rhythm blues and vocal. These music genres will be used for music genre classification which will be done by various machine learning approaches. Various data pre-processing approaches and the various classifiers like Random forest will also be used. The training and testing of data set will be used to find the accuracy of the various music genres. The F-score will also be found. At last the confusion matrix will be plotted for Logistic Regression and other machine learning algorithms.

The total number of audio clips which is taken from Google will be tabulated in every category.

TABLE I. NUMBER OF INSTANCES IN EACH OF GENRE CLASS

	Genre	Count
1	Pop Music	8100
2	Rock Music	7990
3	Hip Hop Music	6958
4	Techno	6885
5	Rhythm Blues	4247
6	Vocal	3363
7	Reggae Music	2997
	Total	**40540**

2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT) | 978-1-6654-6855-8/22/$31.00 ©2022 IEEE | DOI: 10.1109/GCAT55367.2022.9972105

III. VARIOUS DATA PRE-PROCESSING STEPS

Following are the various data pre-processing steps:-

A. Data Pre-processing

An algorithm used in data mining is data pre-processing. It is a process in which raw data is changed in a design that can be understood easily. We cannot work with raw data, A salient step is data pre- processing. Before applying data mining algorithms the quality of data should be checked.

B. Spectrogram Generation

A 2-D description of signal is called spectrogram. A colormap is made use of for quantifying the magnitude of frequency which will be given within a given time window.

A spectrogram changes with time, so it is a visual description of the various frequencies containing spectrum of signal. Spectrograms are infrequently called voicegrams when it is put in an audio signal.

Spectrograms are used widely in the fields of music and seismology.

Amplitude, time and frequency are the three dimensional information of a spectrogram.

C. Deep neural networks

Nodes are little fragments of the system, and they are just like the functioning of neurons present inthe human brain.A process takes place in these nodes,when a stimulus hits them.

A DNN is important when there is a need to subsitute human labour with autonomous work and its efficiency needs to be excluded.There are many applications of DNN in real life. For example, a company which is Chinese naming Sensetime, created a system that consisted of automatic face recognition system which is used to identify criminals by using the real-time cameras in order to find an offender usually in the crowd. Nowadays, it has become a famous practice in the police and various other governmental entities.

IV. LITERATURE REVIEW

A. In the research paper of Content analysis for audio classification and segmentation, Lu L. etal., (2002)

They have put forward their study of classification as well as segmentation of the analysis of audio contents. Here,in this paper a stream of audio is divided according to speaker identity or the audio type. Their dominant proposal is to construct a powerful design which will be efficient of dividing and categorising the audio signals.

B. In the research paper of Automatic musical pattern feature extraction using convolutional neural network, Tom LH .(2010):

They made a huge effort which will be able to know about the various important properties which generally helps to make the requisite design utilized for the classification of music genre,dominant reason of the research area is to recommend a narrative strategy to eliminate various characteristics just like the musical pattern of the audio file by makingthe usage of CNN. The main ambition is to survey the various advantages of CNN in MIR.The full form of MIR is Music Information Retrieval,MIR plays a very salient role in classification of music genre. The experiments

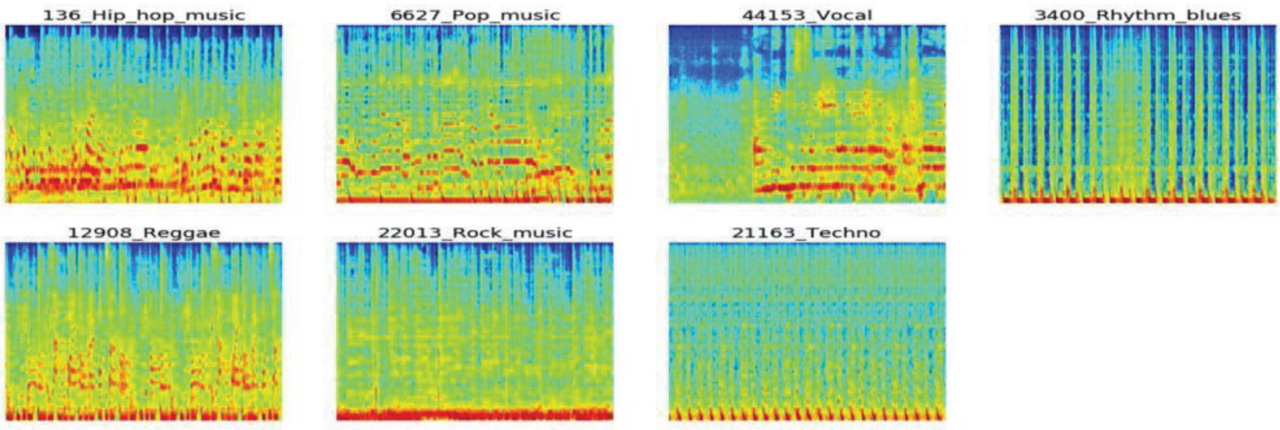

Fig. 1. Sample spectrograms for 1 audio signal from each music genre

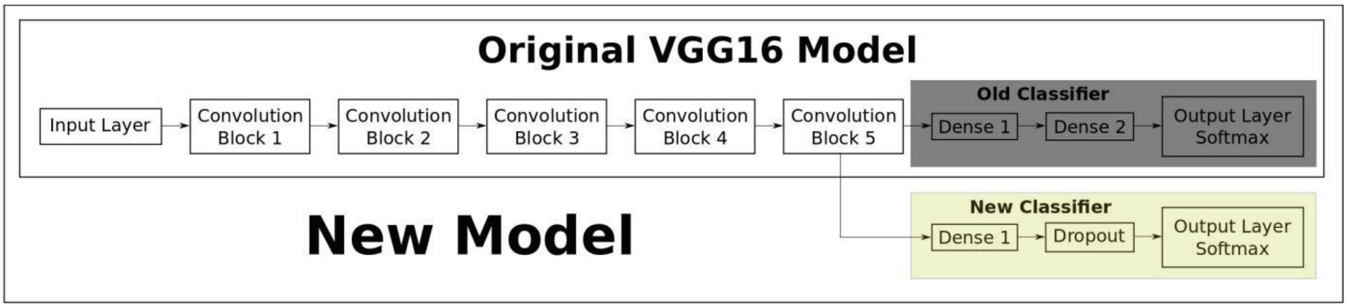

Fig. 2. Convolutional neural network architecture

as well as the results shows us that Convolutional Neural Network have the most powerful ability to catch characteristics like informative characteristics, from the various musical patterns, which are varying.

C. In the research paper of A generative model for raw audio , Aaron Van Den Oord, Nal Kalchbrenner and Koray Kavukcuoglu. 2016.

They gave a brief description of how the audio generation tasks will be carried out. A very common substitute representation is the spectrogram which is present in a signal that captures the details of either frequency or time.

V. RESULTS ANALYSIS AND FUTURE SCOPE

A. Result Analysis

Confusion Matrix for the Extreme Gradient Boosting, relative importance of features in XGBoost model, here the top 20 most contributing features are shown and the models are evaluated. The models are evaluated on the basis of firstly the AUC, secondly accuracy and lastly the F-score.

	Accuracy	F-score	AUC
Spectrogram-based models			
VGG-16 CNN Transfer Learning	0.63	0.61	**0.891**
VGG-16 CNN Fine Tuning	**0.64**	**0.61**	0.889
Feed-forward NN baseline	0.43	0.33	0.759
Feature Engineering based models			
Logistic Regression (LR)	0.53	0.47	0.822
Random Forest (RF)	0.54	0.48	0.840
Support Vector Machines (SVM)	0.57	0.52	0.856
Extreme Gradient Boosting (XGB)	**0.59**	**0.55**	**0.865**
Ensemble Classifiers			
VGG-16 CNN + XGB	**0.65**	**0.62**	**0.894**

Fig. 3. The models are evaluated on the basis of AUC, accuracy and F-score.

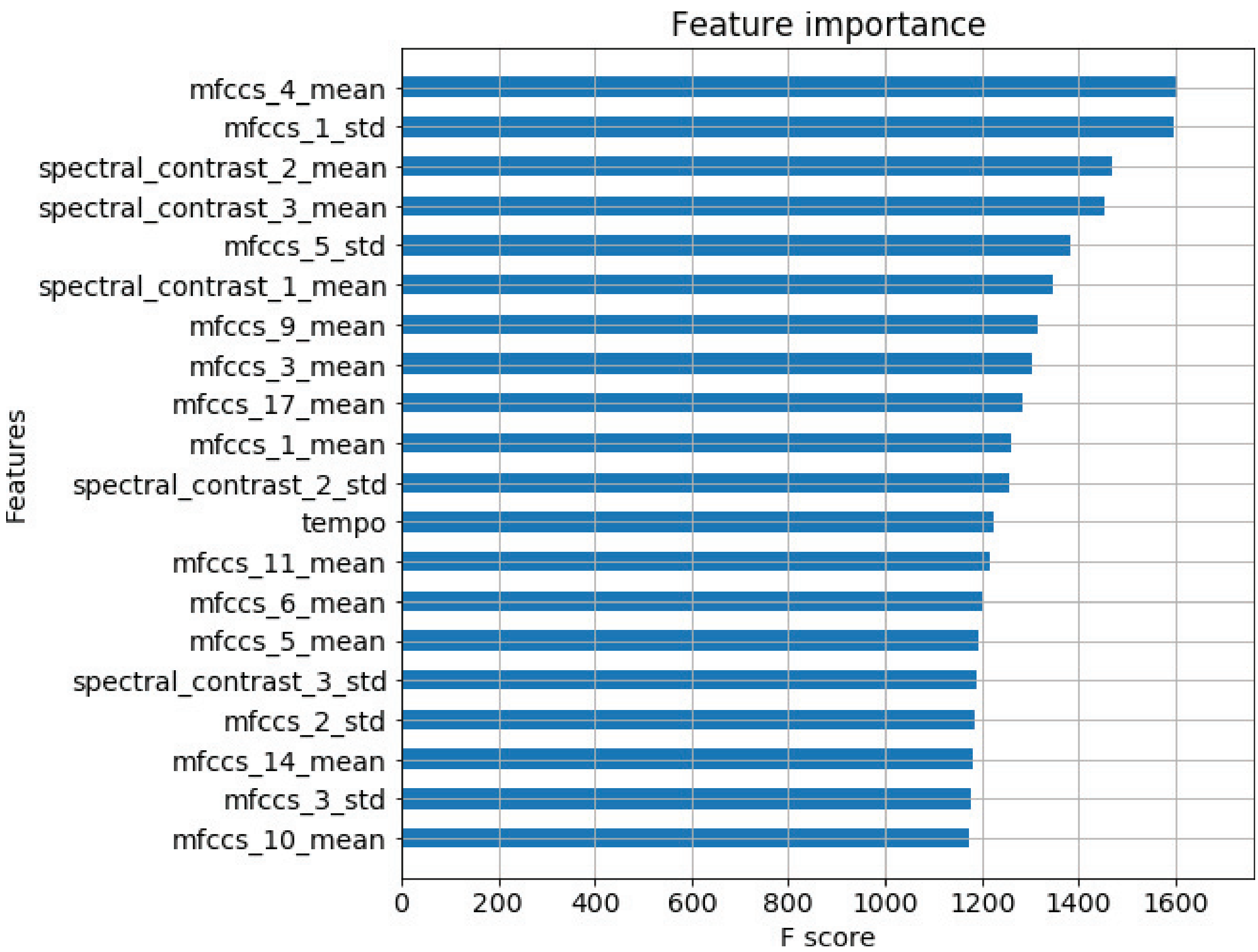

Fig. 4. Relative importance of the features present in XGBoost model,in this figure toptwenty best contributing features are shown

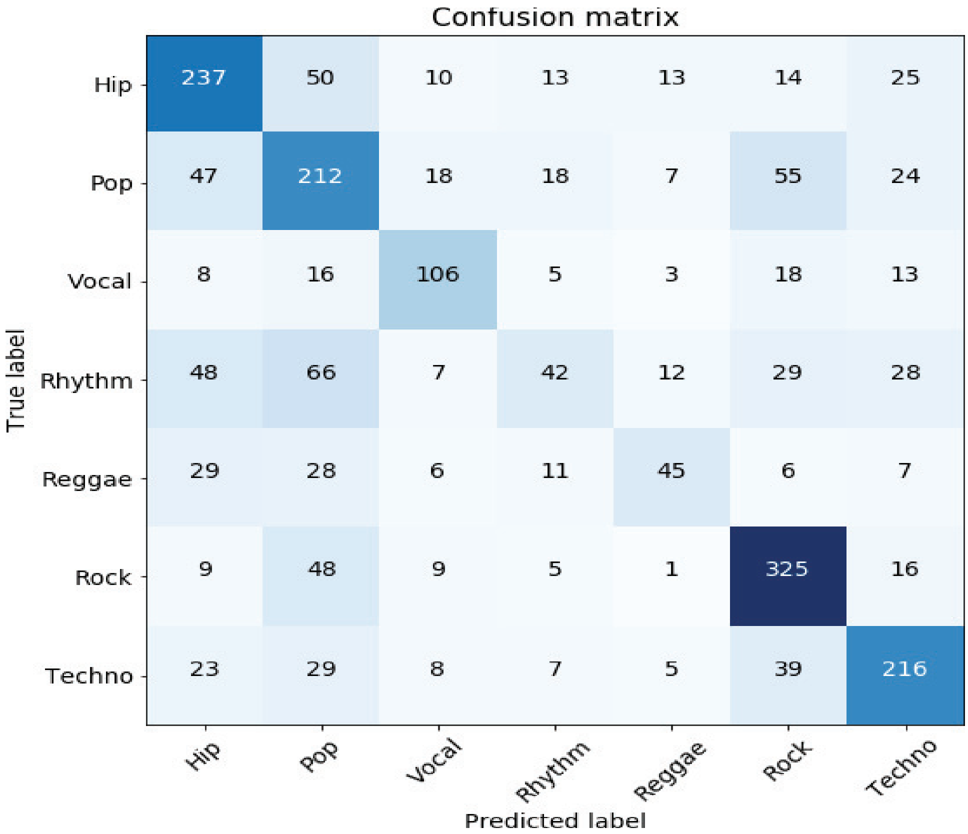

Fig. 5. Confusion Matrix for Extreme Gradient Boosting

VI. FUTURE SCOPE

I started this project as a stepping stone in making a confusion matrix.Extreme Gradient Boosting, relative importance of features in the XGBoost model; the top 20 most contributing features are displayed and the models are evaluated .The models are evaluated on the basis on firstly the AUC,secondly accuracy and lastly the Fscore.

In this work, a technique is used for classification named as Convolutional neural network(CNN),this technique could also train the large amount of dataset and could be split into their various groups according to their features,different classifiers will also be used such as random forest,algorithms like logistic regression,Support Vector Machine and statistical technique like Principal Component Analysis (PCA) and tools like numpy,panda,librosa would be used.

Following are some of the future scopes I think can be listed for now:

1 Music genre classification can be used to produce the confusion matrix for algorithmslike logistic regression.

2 Music genre classification can be used to compare the accuracy of differentmachine learning models.

3 Music genre classification can be used to compare the f-score of various machinelearning algorithms

VII. CONCLUSION

Deep learning approach was the first approach of classification of music genre using machine learning and for the purpose of predicting the genre label of a signal,a CNN model was trained usually from end-to-end, spectrogram was used to carry out this process. Hand-crafted features were used in the second approach which was also the last approach. Four out of many traditional machine learning classifiers were trained beside these features and their performance was compared afterwards. Identification of features that would help the classification of this task was performed. A set data of Audio was used for the experiments and after that an AUC value of 0:894 was reported for an ensemble classifier which merged the two proposed methods. The databases of online music are growing day by day and nowadays it is very simple to access online music, due to this reason people are finding extremely uneasy to regulate the songs that they like to listen to.

Music has nowadays become a significant proportion of Internet content, the most important source of pieces of music is the net. In this context, the automatic procedures which are able to deal with veryhuge amounts of quantities in formats such as digital formats are salient. Labels of music genre are very crucial categories which are used to categorise and classify songs, albums and artists into broader groups which can share alike musical characteristics.

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